

ECEN-629: Homework 5

Due on December 1, 2025 at 11:00 am

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Problem 1

In many cases, we may need to compute the projection of a point onto a convex set (e.g., in projected gradient descent). The projection of x_0 on convex set is defined as the optimal point for the following optimization problem:

$$\begin{aligned} &\text{minimize:} && \|x - x_0\| \\ &\text{subject to:} && x \in \mathcal{C} \end{aligned}$$

This projection is sometimes easy, sometimes hard. For the easy cases, analytical solutions can be found.

- Suppose the norm we use is the ℓ_∞ norm, and the set $\mathcal{C}$ is $\{x \mid \|x\|_2 \leq 1\}$. Find the projection of x_0 .
- Suppose the norm we use is the ℓ_2 norm, and the set $\mathcal{C}$ is $\{x \mid \|x\|_2 \leq 1\}$. Find the projection of x_0 .
- Suppose the norm we use is the ℓ_∞ norm, and the set $\mathcal{C}$ is $\{x \mid l \preceq x \preceq u\}$. Find the projection of x_0 .
- Suppose the norm we use is the ℓ_2 norm, and the set $\mathcal{C}$ is $\{x \mid l \preceq x \preceq u\}$. Find the projection of x_0 .
- Suppose the norm we use is the ℓ_1 norm, and the set $\mathcal{C}$ is $\{x \mid l \preceq x \preceq u\}$. Find the projection of x_0 .
- Suppose the norm we use is the ℓ_2 norm, and the set $\mathcal{C}$ is $\{x \mid Ax = b\}$. Find the projection of x_0 . Assume the matrix A has full row rank, and has more columns than rows.
- Provide one example of a convex set $\mathcal{C}$ that this projection is not easy to obtain analytical solutions or to compute.

Solution

Part (a): Here

$$\begin{aligned} &\text{minimize:} && \|x - x_0\|_\infty \\ &\text{subject to:} && \|x\|_2 \leq 1 \end{aligned}$$

Part (b): Here

$$\begin{aligned} &\text{minimize:} && \|x - x_0\|_2 \\ &\text{subject to:} && \|x\|_2 \leq 1 \end{aligned}$$

Part (c): Here

$$\begin{aligned} &\text{minimize:} && \|x - x_0\|_\infty \\ &\text{subject to:} && l \preceq x \preceq u \end{aligned}$$

Part (d): Here

$$\begin{array}{ll}\text{minimize:} & \|x - x_0\|_2 \\ \text{subject to:} & l \preceq x \preceq u\end{array}$$

Part (e): Here

$$\begin{array}{ll}\text{minimize:} & \|x - x_0\|_1 \\ \text{subject to:} & l \preceq x \preceq u\end{array}$$

Part (f): Here

$$\begin{array}{ll}\text{minimize:} & \|x - x_0\|_2 \\ \text{subject to:} & Ax = b\end{array}$$

Part (g): In the case where $\mathcal{C}$ is an open set.

Problem 2

Consider the problem:

$$\begin{aligned} &\text{minimize: } \text{length}(x) \\ &\text{subject to: } \|Ax - b\| \leq \varepsilon, \end{aligned}$$

where the length function $\text{length}(x) = \min\{k \mid x_i = 0 \text{ for } i > k\}$. The decision variables $x \in \mathbb{R}^n$, and A , b , and ε , are fixed parameters.

- (a) Show the problem is not convex;
- (b) Show the problem is quasiconvex;
- (c) Is the length function equivalent to the sparsity, i.e., the number of non-zero elements of x ?
- (d) (*optional*) Formulate a problem where both the sparsity and the length of x are to be minimized. Propose an approach to solve this problem efficiently.

Solution

Part (a): Let $f(x) = \min\{k \mid x_i = 0 \text{ for } i > k\}$. In order for this function to be convex, we require

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$$

for all x, y and $\theta \in [0, 1]$. That is not the case for this function, hence it is not convex.

Part (b): In order for $f(x)$ to be convex, each sublevel set

$$C_\alpha = \{x \mid f(x) \leq \alpha\}$$

must be convex for all α . In this case, $C_\alpha = \{x \mid \text{length}(x) \leq \alpha\}$. Which defines a convex set for all α . Hence, this problem is quasiconvex.

Part (c): No. This function does not count the number of non-zero elements of x , it only deals with the position of non-zero elements (via index).

Problem 3

Suppose the measurement noise of a scalar quantity x has the truncated Gaussian distribution with parameter $a > 0 > b$, such that the observed measurement y is distributed according to the following distribution

$$p_x(y) = \frac{1}{\Phi\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y-x)^2}{2\sigma^2}\right], \quad \text{for } b \leq y-x \leq a$$

and $p_x(y) = 0$ otherwise, where Φ is a constant, and σ^2 is a fixed and known value.

- Write down an expression for Φ .
- Write down the ML estimation problem, when we have m iid scalar samples $\{y_i\}$, as an optimization problem; simplify it into a standard convex optimization form.
- Suppose $m = 8$, $(y_1, y_2, \dots, y_8) = (-1, 3, -6, 6, 5, 5, 8, 7)$, $\sigma^2 = 1$, and $a = 7$ and $b = -10$. What is the solution to your optimization problem?

Solution

Part (a):

$$\int_{b+x}^{a+x} \frac{1}{\Phi\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y-x)^2}{2\sigma^2}\right] dy = 1$$

Part (b): For iid samples,

$$l(x) = \sum_{i=1}^m \ln(1) - \ln(\Phi\sqrt{2\pi\sigma^2}) - \frac{(y_i - x)^2}{2\sigma^2}$$

Which turns into

$$\begin{aligned} \text{minimize:} \quad & -\sum_{i=1}^m (y_i - x)^2 \\ \text{subject to:} \quad & b \leq y_i - x \leq a, \quad i = 1, \dots, m \end{aligned}$$

Part (c): Now $m = 8$, $(y_1, y_2, \dots, y_8) = (-1, 3, -6, 6, 5, 5, 8, 7)$, $\sigma^2 = 1$, and $a = 7$ and $b = -10$.

Problem 4

Write a simple supporting vector machine program using Matlab/CVX/Python as follows:

- (a) Write a function to generate $n = 800$ random points in the dimensional space from the Gaussian mixture distribution

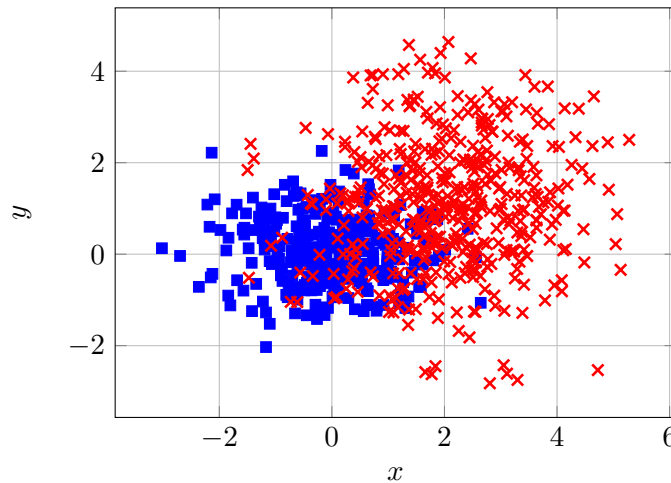
$$\begin{aligned}
 P(x \mid \mu_{1,x}, \mu_{1,y}, \sigma_{1,x}^2, \sigma_{1,y}^2, \pi_1, \mu_{2,x}, \mu_{2,y}, \sigma_{2,x}^2, \sigma_{2,y}^2, \pi_2) \\
 = \frac{\pi_1}{2\pi\sigma_{1,x}\sigma_{1,y}} \exp\left(-\frac{(x - \mu_{1,x})^2}{2\sigma_{1,x}^2} - \frac{(y - \mu_{1,y})^2}{2\sigma_{1,y}^2}\right) \\
 + \frac{\pi_2}{2\pi\sigma_{2,x}\sigma_{2,y}} \exp\left(-\frac{(x - \mu_{2,x})^2}{2\sigma_{2,x}^2} - \frac{(y - \mu_{2,y})^2}{2\sigma_{2,y}^2}\right)
 \end{aligned}$$

where $(\mu_{1,x}, \mu_{1,y}) = (0, 0)$, $(\mu_{2,x}, \mu_{2,y}) = (2, 1)$, $(\sigma_{1,x}^2, \sigma_{1,y}^2) = (1, 0.5)$, $(\sigma_{2,x}^2, \sigma_{2,y}^2) = (\sqrt{2}, 2)$, $\pi_1 = 0.4$ and $\pi_2 = 0.6$. In other words, in this mixture Gaussian distribution, the modes are chosen at $(0.4, 0.6)$ probability. Draw the scatter plot of these points, and label each point using either a square or a cross, depending on which mode (-1 or 1) the data point was produced from.

- (b) Write a program to call an optimization solver to solve the dual optimization problem, to find the support vector classifier using 600 samples, and plot the classifier and on the scatter plot of these 600 samples.
- (c) Classify the remaining 200 (holdout) points using the classifier you obtained from SVM; plot (in a new figure) the 200 points and classifier result. What is the error rate?
- (d) Complete the task in (b) and (c) using existing Python/Matlab functions or packages. Compare to the result you obtained.

Solution

Part (a):



Problem 5

Let $\gamma > 1$. The function

$$f(x_1, x_2) = \begin{cases} \sqrt{x_1^2 + \gamma x_2^2} & |x_2| \leq x_1; \\ \frac{x_1 + \gamma|x_2|}{\sqrt{1 + \gamma}} & \text{otherwise.} \end{cases}$$

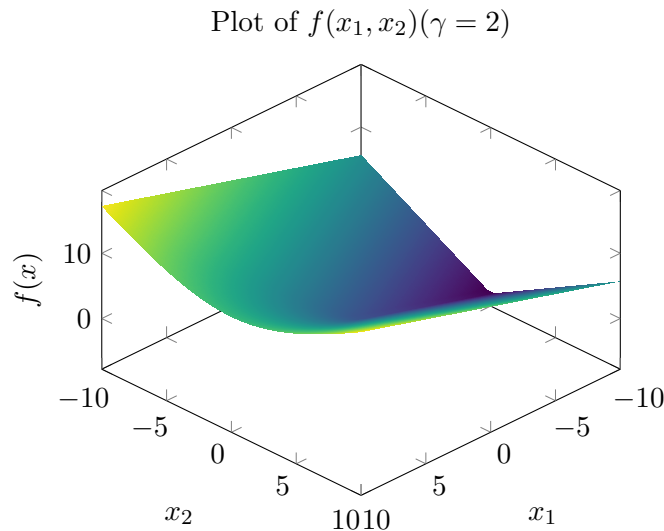
is convex.

- Plot the function and visually identify (approximately) its optimal solution;
- Let $\gamma = 100$. Let $x^{(k)} = (1, 5)$ be the point in the k -th iteration. Find the search direction of the gradient descent method.
- Again let $\gamma = 100$, and let $x^{(k)} = (1, 5)$ be the point in the k -th iteration. Find the search direction of the steepest descent method under the ℓ_1 norm;
- Again let $\gamma = 100$, and let $x^{(k)} = (1, 5)$ be the point in the k -th iteration. With the gradient descent method and backtracking line search of parameters $(\alpha, \beta) = (0.45, 0.9)$, will the step size be determined in two iterations in the line search procedure?
- Applying the gradient descent algorithm on this function, with starting point $x^{(0)} = (\gamma, 1)$ with an exact line search. It can be shown that the iterations for $x_1^{(k)}$ is

$$x_1^{(k)} = \gamma \left(\frac{\gamma - 1}{\gamma + 1} \right)^k.$$

Find the value $x_2^{(k)}$ in that corresponding iteration. What does the algorithm converge to? Is it optimal?

Solution



Problem 6

Consider the optimization problem

$$\begin{aligned} \text{minimize:} \quad & \log \left(\sum_{i=1}^n i e^{x_i} \right) + \sum_{i=1}^n i^2 (x_i - i)^2 \\ \text{subject to:} \quad & \sum_{i=1}^n x_i = P. \end{aligned}$$

- Let $P = 15$. Assume $n = 5$. If the starting point is $x^{(0)} = (1, 2, 3, 4, 5)$, find the gradient, the Hessian matrix, and the first Newton step. Write down the equation, but you can solve for it numerically.
- Let $P = 15$. Assume $n = 50$. Write your own Newton method program (in c/c++ or python) with infeasible start to calculate the optimal solution.
- Use CVX package to solve the same problem in (b).

Solution

The gradient

$$\nabla f = \frac{i e^{x_i}}{\sum_j j e^{x_j}} + 2i^2 (x_i - i)$$

The Hessian

$$\nabla^2 f = \begin{bmatrix} \frac{e^{x_1}}{\sum_k k e^{x_k}} - \left(\frac{e^{x_1}}{\sum_k k e^{x_k}} \right)^2 + 2 & -\frac{1e^{x_2} 1e^{x_2}}{\sum_k k e^{x_k}} & -\frac{1e^{x_1} 3e^{x_3}}{\sum_k k e^{x_k}} \\ -\frac{2e^{x_2} 1e^{x_1}}{\sum_k k e^{x_k}} & \frac{4e^{x_2}}{\sum_k k e^{x_k}} - \left(\frac{2e^{x_2}}{\sum_k k e^{x_k}} \right)^2 + 8 & -\frac{2e^{x_2} 3e^{x_3}}{\sum_k k e^{x_k}} \\ -\frac{3e^{x_3} 1e^{x_1}}{\sum_k k e^{x_k}} & -\frac{3e^{x_3} 2e^{x_2}}{\sum_k k e^{x_k}} & \frac{9e^{x_3}}{\sum_k k e^{x_k}} - \left(\frac{3e^{x_3}}{\sum_k k e^{x_k}} \right)^2 + 18 \end{bmatrix}$$

With $x_1 + x_2 + x_3 + x_4 + x_5 = 15$.

Problem 7

Consider the optimization problem (similarly as above but with an inequality constraint)

$$\begin{aligned} \text{minimize:} \quad & \log \left(\sum_{i=1}^n i e^{x_i} \right) + \sum_{i=1}^n i^2 (x_i - i)^2 \\ \text{subject to:} \quad & \sum_{i=1}^n x_i \leq P. \end{aligned}$$

- (a) Let $P = 20$. Assume $n = 5$. If the starting point is $x^{(0)} = (1, 2, 3, 4, 5)$. Find the gradient and the Hessian of the logarithmic barrier function.
- (b) Let $P = 20$. Assume $n = 5$. If the starting point is $x^{(0)} = (1, 2, 3, 4, 5)$. In the log barrier method (when $t = 10$), find the first Newton step. Write down the equation, but you can solve for it numerically.
- (c) Use CVX package to solve the same problem in (b).

Solution

The gradient

$$\nabla f = \frac{i e^{x_i}}{\sum_j j e^{x_j}} + 2i^2(x_i - i)$$

The Hessian

$$\nabla^2 f = \begin{bmatrix} \frac{e^{x_1}}{\sum_k k e^{x_k}} - \left(\frac{e^{x_1}}{\sum_k k e^{x_k}} \right)^2 + 2 & -\frac{1e^{x_2} 1e^{x_2}}{\sum_k k e^{x_k}} & -\frac{1e^{x_1} 3e^{x_3}}{\sum_k k e^{x_k}} \\ -\frac{2e^{x_2} 1e^{x_1}}{\sum_k k e^{x_k}} & \frac{4e^{x_2}}{\sum_k k e^{x_k}} - \left(\frac{2e^{x_2}}{\sum_k k e^{x_k}} \right)^2 + 8 & -\frac{2e^{x_2} 3e^{x_3}}{\sum_k k e^{x_k}} \\ -\frac{3e^{x_3} 1e^{x_1}}{\sum_k k e^{x_k}} & -\frac{3e^{x_3} 2e^{x_2}}{\sum_k k e^{x_k}} & \frac{9e^{x_3}}{\sum_k k e^{x_k}} - \left(\frac{3e^{x_3}}{\sum_k k e^{x_k}} \right)^2 + 18 \end{bmatrix}$$

With $x_1 + x_2 + x_3 + x_4 + x_5 \leq 15$.

Problem 8

Consider the unconstrained optimization problem

$$\text{minimize: } e^{x_1+2x_2} + x_2^2 + x_3^2 - x_2x_3$$

where $x = (x_1, x_2, x_3)^T \in \mathbb{R}^3$ is the variable. Let $x^{(k)} = (1, 1, 1)$ be the point in the k -th iteration.

- (a) Compute the next Newton step.
- (b) Suppose the optimization problem has two additional constraints:

$$x_1 + x_2 + x_3 \leq 4$$

$$x_2 - x_3 \leq 2.$$

Compute the next Newton step using the log-barrier method with the same $x^{(k)}$ and the barrier weighting parameter $t = 20$.

Solution

Part (a): The Newton step

$$\Delta x_{\text{nt}} = -\nabla^2 f(x)^{-1} \nabla f(x)$$

with

$$\nabla f(x) = \begin{bmatrix} e^{x_1+2x_2} \\ 2e^{x_1+2x_2} + 2x_2 - x_3 \\ x_2 + 2x_3 \end{bmatrix}$$

and

$$\nabla^2 f(x) = \begin{bmatrix} e^{x_1+2x_2} & 2e^{x_1+2x_2} & 0 \\ 2e^{x_1+2x_2} & 2e^{x_1+2x_2} + 2 & -1 \\ 0 & 1 & 2 \end{bmatrix}$$

Part (b):

$$\text{minimize: } e^{x_1+2x_2} + x_2^2 + x_3^2 - x_2x_3$$

$$\text{subject to: } x_1 + x_2 + x_3 \leq 4$$

$$x_2 - x_3 \leq 2$$